MOBILE APPLICATION DEVELOPMENT (MAD)

FitHydro Tracker

Smart Hydration & Fitness Companion App

Built with Native Android (Kotlin) • SQLite Database • ViewBinding

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1. Introduction & Motivation

The Modern Lifestyle Challenge

• Water constitutes approximately 60-70% of human body weight and is critical for cognitive and physical performance.

• Due to prolonged desk work and screen time, over 75% of individuals suffer from chronic mild dehydration.

• Physical exercise and daily water intake are inextricably linked: sweat loss must be balanced with targeted water consumption.

• Users frequently struggle to form sustainable hydration habits without consistent feedback and micro-rewards.

Project Motivation & Purpose

• Empower users with a unified, lightweight mobile tracker for both water intake and active workout calories.

• 100% Privacy-First & Offline: No accounts, no internet connection, and zero ads required.

• Instant One-Tap Logging: Minimize friction so logging a drink takes less than 2 seconds.

• Demonstrate core MAD engineering principles: SQLite persistence, lifecycle management, animations, and clean UI architecture.

2. Problem Statement & Key Objectives

Problem Statement

• Existing commercial fitness and water apps are overly bloated, requiring cloud logins and subscriptions.

• They frequently sell user health metrics or bombard users with intrusive pop-up ads.

• Most apps treat water and fitness as separate silos, failing to correlate hydration requirements with workout exertion.

• Heavy background sync consumes battery, RAM, and mobile data unnecessarily.

Core Project Objectives

• Develop an ultra-fast Native Android application in Kotlin using modern Material 3 design.

• Provide quick-add presets (250ml glass, 500ml bottle, 750ml flask) and custom logging.

• Implement sports-science dynamic hydration calculator based on user weight and activity level.

• Include multi-exercise workout logging with automatic calorie burn formula calculations.

• Ensure resilient offline persistence with SQLite CRUD operations and SharedPreferences.

3. Literature Survey & Comparative Analysis

Existing Commercial Apps

• MyFitnessPal / Waterllama

• • Pros: Rich graphics, social sync.

• • Cons: Paywalled premium tiers ($30+/yr), mandatory internet, excessive telemetry.

• • High APK size (>80 MB) and background RAM overhead.

Basic Academic Demos

• Standard College Lab Projects

• • Pros: Simple to write.

• • Cons: Unstyled default XML buttons, static hardcoded values, lack of animations.

• • No database CRUD or streak retention across app restarts.

Proposed FitHydro Solution

• FitHydro Tracker Advantages

• • Native Kotlin with smooth ObjectAnimator progress rings.

• • Integrated SQLite CRUD + SharedPreferences.

• • Sports-science adaptive water calculator.

• • Lightweight (<6MB APK), 0 permissions except haptic vibration, 100% offline.

4. System Requirements & Technology Stack

Software & Development Stack

• Language: Kotlin 1.9.22 (Modern, null-safe, concise).

• Target Platform: Android 14 (API 34) | Minimum: Android 7.0 (API 24).

• UI Framework: Material 3 Design Components + ConstraintLayout.

• Database: Android SQLite (`SQLiteOpenHelper`) for persistent CRUD.

• Preferences: Android SharedPreferences for streaks & user goals.

• Build System: Gradle 8.2 & Android Gradle Plugin (AGP) 8.2.0.

Hardware & Execution Environment

• Development PC Requirements:

• • Windows 10/11, macOS, or Linux.

• • 8GB+ RAM, Java Development Kit (JDK 17/20).

• • Android Studio Iguana / Hedgehog / Koala.

• Target Mobile Device Requirements:

• • Any Android phone or emulator running Android 7.0+.

• • Minimum 1 GB RAM, only ~10 MB storage required.

5. System Architecture & Data Flow

Architectural Layers

- Presentation Layer (UI):
- • `MainActivity.kt` orchestrating views, dialogs, and animations.
- • ViewBinding for direct, null-safe XML widget binding.
- Business Logic Layer:
- • Hydration formula engine (`Weight × 35ml + Activity Offset`).
- • Workout calorie calculator (MET rates per exercise type).
- • Consecutive day streak calculator (`SimpleDateFormat`).
- Data Persistence Layer:
- • SQLite `daily_logs` table + SharedPreferences.

Data Flow Pipeline

- 1. User Action: Tap quick-add button or submit dialog.
- 2. Event Handling: Click listener triggers haptic vibration (`Vibrator`).
- 3. Calculation: Compute intake %, calories burned, and date string.
- 4. Storage: Execute SQLite `insertLog()` or update SharedPreferences.
- 5. UI Animation: `ObjectAnimator` sweeps the circular progress ring from previous to new percentage.
- 6. Feed Update: RecyclerView adapter re-queries today's records and refreshes view.

6. Database Design & Local Storage

SQLite Database Schema (`fithydro.db`)

• Table Name: `daily_logs`

• • `id`: INTEGER PRIMARY KEY AUTOINCREMENT

• • `type`: TEXT (Category: 'WATER' or 'WORKOUT')

• • `title`: TEXT (e.g. 'Glass of Water', 'Running (30 mins)')

• • `detail`: TEXT (e.g. '+250 ml', '+300 kcal')

• • `amount`: INTEGER (Numeric ml or kcal for aggregations)

• • `timestamp`: INTEGER (Epoch millis for sorting)

• • `date_string`: TEXT (Format 'YYYY-MM-DD' for date queries)

SharedPreferences Key-Value Storage

• File: `fit_hydro_prefs.xml`

• • `water_goal` (Int): User daily water intake target (Default 2500 ml).

• • `calorie_goal` (Int): User active burn target (Default 400 kcal).

• • `weight_kg` (Int): User body weight for calculator (Default 70 kg).

• • `activity_level` (Int): 0 (Low), 1 (Moderate), 2 (High).

• • `streak_count` (Int): Number of consecutive logging days.

• • `last_streak_date` (String): Last recorded calendar date.

7. Core Features & Technical Highlights

Dynamic Animated Progress Ring

• Custom Layer-List XML ring with sweep gradient (#00B4D8 to #0077B6).

• Smooth hardware-accelerated transitions via `ObjectAnimator`.

One-Tap Rapid Quick Logging

• Quick tap tiles: 250ml Glass, 500ml Bottle, 750ml Flask, & Custom Dialog.

• Haptic vibration response (`VibrationEffect`) for tactile feedback.

Scientific Hydration Calculator

• Sports science formula: `Base (kg * 35ml) + Activity Intensity Factor`.

• 1-Tap 'Apply as Goal' button sets the calculated value as active target.

Multi-Exercise Fitness Logger

• 6 Activities: Running (10 kcal/min), Cycling (8), Gym (6), Walking (4), Swimming (9), Yoga (3).

• Real-time calorie prediction as the user inputs duration.

8. Testing & Test Cases

Executed Verification Test Cases

• TC-01 [Water Quick Log]: Tap +250ml button -> Entry written to SQLite, progress ring advances by 10%, haptic feedback triggers. [STATUS: PASS]

• TC-02 [Custom Input Validation]: Enter non-numeric or empty string in Custom Dialog -> System shows validation toast, rejects invalid state. [STATUS: PASS]

• TC-03 [Workout Calorie Calculation]: Select 'Running' for 30 minutes -> Validated 300 kcal burned, progress bar updates proportionally. [STATUS: PASS]

• TC-04 [Data Persistence Across Process Death]: Add logs, swipe kill app from Recent Apps, reopen -> All logs, total ml, and streaks restored. [STATUS: PASS]

• TC-05 [Log Deletion & Auto-Recalculate]: Click trash icon on 500ml log -> Log deleted from SQLite, ring decreases smoothly by 20%. [STATUS: PASS]

• TC-06 [Screen Rotation & Density]: Tested across different DPIs (mdpi to xxxhdpi) -> ConstraintLayout maintains proportional layout without clipping. [STATUS: PASS]

9. User Interface & Demonstration Flow

Visual Hierarchy & Material 3 UI

• Top Header: Gradient navy surface displaying real-time date, 🔥 Streak badge, and top shortcut chips.

• Hydration Dashboard: Centered circular progress with percentage, remaining target counter, and motivational status text.

• Interactive Grid: Modern elevated cards with emoji iconography for rapid drinking intake.

• Fitness Summary Card: Calorie burn progress bar with distinct vibrant coral accenting.

• Activity Feed: Chronological RecyclerView cards with type indicators and individual delete actions.

Faculty Demo Walkthrough Steps

• 1. Open App: Point out current date and 1-Day Streak indicator.

• 2. Tap Glass (250ml): Demonstrate circular progress animation & haptic tap.

• 3. Log 30m Running: Show dynamic calorie calculation and fitness progress.

• 4. Open Calculator: Show dynamic formula adjusting goal based on weight.

• 5. Delete Entry: Show real-time SQLite deletion and UI update.

• 6. Force Close & Reopen: Prove 100% offline persistence.

10. Conclusion & Future Scope

Project Achievements & Conclusion

• Successfully developed and deployed a native Android mobile application adhering to MAD syllabus requirements.

• Demonstrated proficiency in Kotlin, Material 3 UI, SQLite databases, SharedPreferences, and ViewBinding.

• Built a genuinely useful, privacy-first, zero-bloat daily utility that runs completely offline with zero latency.

• Achieved a highly optimized build: APK size is under 6MB and cold boot time is under 800ms.

Future Enhancements

• Periodic Smart Drink Alarms: Implementing Android `WorkManager` for battery-efficient background hydration reminders.

• Historical Analytics & Visual Charts: Integrating `MPAndroidChart` for weekly/monthly trend graphs.

• Health Connect & Google Fit API: Syncing step counts and automated calorie imports from smartwatches.

• Export / Backup: Export logs to CSV/PDF reports for doctor/trainer consultation.